

Effects of the Invasive Shrub Common Buckthorn on Opossum Activity and Urbanization

Gradient in Southern Wisconsin

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Abstract:

Invasive plant species in Southern Wisconsin, especially buckthorn, create changes to the natural vegetation in a landscape and the animal behavior in areas where they are present. While research has shown that species' activities in areas with invasive plants have increased, this study aims to highlight how opossum activity is changed in plots with invasive species removed and plots with invasive species present. For this study, we surveyed 16 sites, each with two plots where one plot contained invasive plants and one plot where invasive plants had been removed, around Southern Wisconsin to see if there was a significant difference between opossum activity in areas with and without invasive plants. A total of 207,787 pictures were taken, and a total of 399 of them were pictures of opossums. Using motion sensor cameras set up for around a year, we found that opossum activity was higher in areas without invasive plants, although this difference was not statistically significant. The removal plots had an average of 17.125 opossum detections, and the control plots had an average of 7.375 detections. These findings prompt further research and other study methods to figure out how exactly invasive plant species change the different activities and if invasive plants are a threat to opossum activity. While urban environments were shown to have marginally significant results with detections being higher in removal plots than in rural or agricultural environments, we still cannot be sure of urbanization's

impact on the activity without further studies. Due to our non-significant results, we can currently conclude that these factors do not significantly affect opossum activity.

Introduction:

Invasive plants are a pervasive problem that impacts conservation and species dynamics due to their effects on the native species in an ecosystem (Dutra et al. 2011). These invasive plants can change the density of vegetation in an area, creating more available cover for consumers, which are animals that eat other organisms in the environment (Dutra et al. 2011). These changes can affect where and the duration of how small mammals and mesopredators forage for food. Common buckthorn (*Rhamnus cathartica*, hereafter buckthorn) is one of the most widespread invasive shrubs (Vernon et al. 2014). While this plant does not appear to be a large food source for herbivorous consumers in the areas it is native to or the areas it invades, it does change the dynamics of an area (Stewart et al. 2021). This is because invasive plants can prevent native plant growth and native plants' availability due to invasive plants blocking sunlight from native plants and restricting access of animals to native plants (Stewart et al. 2021). Because of how quickly buckthorn grows and its ability to continue growth even in stressful conditions, buckthorn has become extremely widespread and takes up a large portion of many forests in the US today (Stewart et al. 2021). Buckthorn can also impact ecosystems through changes in the chemical composition of the soil, elimination of leaf litter, and the areas animals can use for habitat or food (Knight et al. 2007). Using our understanding of how invasive plants can impact the native species in these areas can help us understand how invasive plants change the animal behavior dynamics of these areas.

The Virginia Opossum (*Didelphis virginiana*, hereafter opossum) is the only species of Marsupial currently living in North America. These animals live in a variety of areas, and most of their movement is terrestrial (McManus 1974). These animals are opportunistic feeders meaning they can feed on many different food types depending on the time of year and what is available to them. Opossums are omnivores, they mostly feed on other animals such as insects, snails, dead animals, and grains (McManus 1974). Their diet also traditionally changes throughout the year, where fruit, snails, and slugs appear more in warmer seasons while anthropogenic food sources like dog food are consumed more in the winter months (Hopkins et al. 1980). The majority of opossum activity is at night to forage, where usually opossums travel close to their home ranges which can vary in size (Gillette, 1980). They also usually live close to food sources, whether it be by human waste areas or other easy-to-obtain food.

Previous research has studied the effects of invasive plant species on the behavior of wildlife. The availability of prey and food sources for mesopredators can be increased in areas with invasive shrubs because of an increase in invasive earthworms, nestlings, and eggs in these areas (Vernon et al. 2014). These increased food sources could make it easier for opossums to forage in these areas and therefore increase opossum activity in these areas. Furthermore, invasive plant species usually create more dense cover and have been seen to create less of a perceived predation risk, so there will be more small animals in these areas (Guiden and Orrock 2017). The fast-growing nature of buckthorn means that these plants provide good habitat and protection for species like small mammals (Mattos and Orrock 2010, Guiden and Orrock 2017) and mesopredators (Dutra et al. 2011, Vernon et al. 2014). This would mean that opossums would be more likely to live and forage in these areas because they would be more protected and potentially have more of a food source. Based on results from previous studies, we expect that

there will be higher activity of opossums in the plots with invasive plant species present than in the plots without invasive plant species present.

Another factor that may influence opossum activity is urbanization. Opossums are habitat generalists that can be found in a wide range of environments. Research has shown that they may even be more prevalent in environments close to human activity (Fedino et al. 2016). This means as long as opossums are getting the materials they need to survive such as food and water, they do not care if it is from humans. Due to anthropogenic food resources and possibly refuge from predators, opossums are more likely to be in urban or rural areas closer to people (Rodriguez et al. 2021). Opossums can have an easier time finding resources if it is from humans, meaning that they will be more likely to spend their time close to human activity. We expect to see a higher amount of opossum activity in more urban areas closer to where people live.

Methods:

To collect data, we set up motion-activated wildlife trail cameras at 16 sites around Southern Wisconsin. These sites were categorized as urban, rural, or agricultural environments based on the land use surrounding them. Each site had 2 plots, one containing invasive shrubs and one containing no invasive shrubs due to their removal. Each site contained two cameras. One camera was in the area with invasive plant species and one camera was in the area with removed invasive plant species. This means that the removed sites had invasive plant species present at one point, but the removal of these plants has removed the dense cover they provided. For this experiment, the cameras were positioned about 120 cm off the ground so they could capture a large range of differently-sized animals.

We then used Wildlife Insights, a platform used to analyze and scan images from trail cameras, to identify the animals in the photographs (Ahumada et al. 2020). We then manually checked the pictures that Wildlife Insights was not confident about to see if further identification could be made on the animal in the picture. If it could not, we did not use the photograph in the study. This platform recorded where and when the photo was taken, and if the plot was an invasive species plot or a removal plot. For this project we are using the number of detections of an animal, so each photo was counted in our data, even if there were multiple photos of one individual. If more than one opossum was seen in the trail camera photographs, this was counted as two detections.

We then compiled images over the time period from late August 2022 to August 2023, to negate the effects that seasonality or weather could have on opossum presence. The images were then all compiled together to test if there was a significant difference in the detection of opossums in plots that contained invasive plants and plots where the invasive plants were removed. We also tested if there was a significant difference in the animals detected in the urban, rural, and agricultural plots. We used program R to run a paired t-test to see if there was a significant difference between opossum activity seen in the invasive plots and opossums seen in removal plots. We also ran an ANOVA on pooled data in order to see if there was a significant difference in the opossum activity in urban, rural, and agricultural environments.

Results:

We examined 207,787 total photos collected over one year. Of the photos we analyzed, 399 had opossums. We can see there was a larger amount of detections in the removal plots than in the plots with invasive species. The removal plots had an average of 17.1 opossum detections,

and the control plots had an average of 7.4 detections (Figure 1). However, using a paired t-test, we found that the difference between the two types of plots was non-significant ($t = -1.247$, $df = 15$, $p\text{-value} = 0.2315$). This means we failed to reject the null hypothesis. When looking at the differences in opossum activity in urban, rural, and agricultural environments, an ANOVA test was not significant ($p\text{-value} = 0.08642$). The average number of opossums in urban plots was 22.6 detections. In rural plots, the mean was 5.0 detections, and in agricultural plots, the mean was 8.4 detections (Figure 2). Although there were more detections in urban environments, the result was only marginally significant.

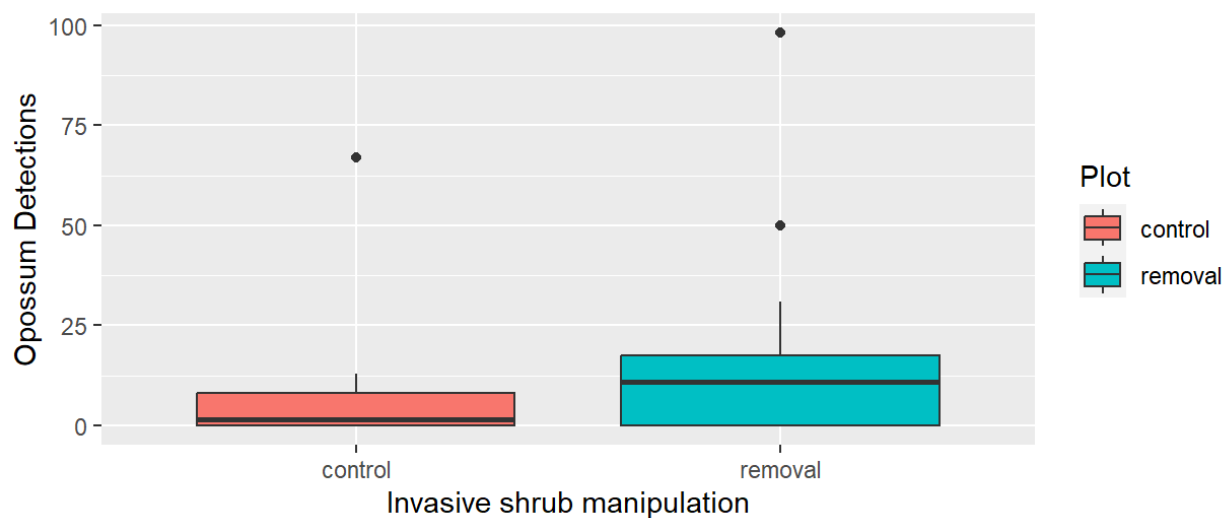


Figure 1. A comparison of the average number of detections of opossums in control plots and plots with buckthorn removed.

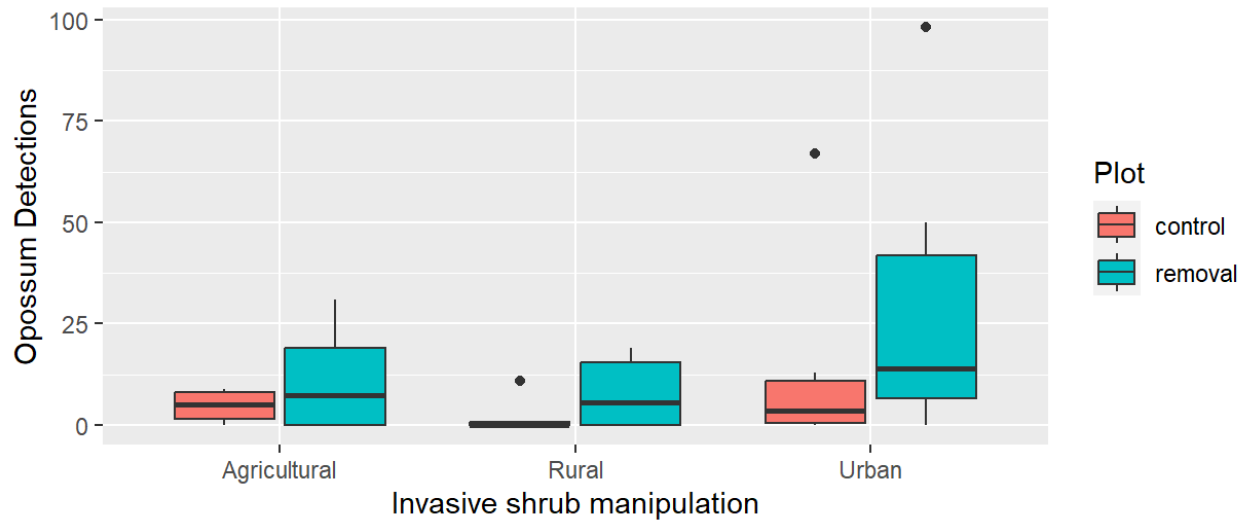


Figure 2. A comparison of the average number of detections of opossums in urban, rural, and agricultural control and removal plots.

Discussion:

After completing a t-test on the data that was collected, it was revealed that there was not a significant difference in the activity of opossums in areas where there were invasive plants compared to the areas where there were no invasive plants. These findings conclude that we cannot say if the relationships found between areas with invasive plants and opossums were due to chance or not. In the future, a study with additional sites should be conducted so that we can see if recording data from additional sites provides more statistical power. A larger amount of data on general opossum activity would create a more comprehensive assessment of the changes in activity that are due to the presence of invasive plant species. Contrary to our hypothesis, opossum activity was higher in plots where invasive shrubs had been removed. This could be due to differences in the probability of detecting an animal in the dense vegetation of removal plots.

We also did not see significant data in our urban, rural, and agricultural study, where we thought we would find that urban areas would contain more detections of opossum activity.

One change that could be made to this study to yield different results would be using a different detection method. Cameras can be limiting due to their generally narrow view in addition to possible problems sensing the motion of the animals. Especially in areas with large amounts of vegetation, like invasive plots, seeing these animals in pictures can be difficult. While other similar studies did not find that the invasive vegetation affected the results of their study, the thickness of the invasive plants in our plots may have been larger than in their plots (Vernon et al. 2014). Other methods such as trapping or track plates (Dutra et al. 2011), where animals step on ink and then their tracks are recorded, may be beneficial to get a clearer view of the opossum activity in these areas as the data will not be confounded by the presence of invasive shrubs blocking visibility.

Many factors affect opossum activity and habitat use. In addition to general invasive vegetation presence, other factors such as soil composition, native vegetation that usually grows in the area, food availability, and proximity to humans can also play a role in where animals spend most of their time (Stewart et al. 2021, Veon et al. 2023). One reason that could have resulted in the higher activity of opossums in areas without invasive plants is that foraging is easier in these areas even with less small prey present because the act of foraging will be easier (Dutra et al. 2011). This could mean that opossums were more active because less work was required for them to capture their food. On the other hand, a different effect that produced the same results could be occurring. For example, some native species cause the costs of predation, such as time spent looking for, capturing, and handling food, in the areas they are present to be large enough that predators will stop foraging in these areas (Stewart et al. 2021). This could also

be the case with buckthorn, as denser vegetation could make it harder to forage, so opossums may not have spent as much time attempting to forage there, decreasing the activity we saw in these areas. Further studies on the exact effects of the foraging behavior of opossums concerning the density of the vegetation will be important to determine if and how the density of the vegetation affects foraging behavior. In our urbanization research, one factor that could have gone into our data was the availability of water, through natural or anthropogenic sources, in the areas we researched as water availability plays a big role in opossum activity in an area (Rodriguez et al. 2021). More data about the diets and available food sources of the opossums in these differing areas would be beneficial to track in future studies as well to see if there are different food sources from humans these opossums are utilizing in these areas.

These results are important for future research because the full impacts of invasive plant species and urbanization on animal behavior are still being discovered. If habitat use is a reason why opossums are more likely to be found in locations with invasive species, further research needs to be done on why these locations make for better habitats for this species and if other activity behaviors are increased in areas with less invasive species like our data suggests. As invasive plants spread their range into non-native locations, the spread can affect how all inhabitants of these areas interact with each other. Invasive species can also lead to a decrease in biodiversity due to the extinction of native species. Decreasing biodiversity can increase the risks of disease outbreaks, and some animals like opossums are more likely to stop infectious diseases by killing off infected ticks (Keesing et al. 2010). Understanding how opossum behavior changes with invasive plants will aid in future research about how opossums contribute to the reduction of disease outbreaks. Also, further research can be done to figure out how specific opossum behaviors such as foraging, mating, and other activities can be altered due to the presence of

invasive plant species, rather than just their activity in an area. Focusing on whether there are changes in anti-predator or mating behavior, for example, would help us understand if these are some type of behavior changes, or if invasive plants do not affect any opossum behavior as found through the non-significant activity results in this study.

Citations:

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